Data Structures - Important Questions

UNIT-1: Structures, Unions & Introduction to Data Structures

1. Define Structure. Explain structure declaration and initialization with an example.
2. Explain Nested Structures with a suitable C program.
3. Explain Self-Referential Structures with an example.
4. Differentiate Structure and Union.
5. Explain Enumerated Data Type (enum) with example.
6. Explain typedef keyword with suitable example.
7. Explain Bit Fields and their advantages.
8. Define Data Structure. Explain types of Data Structures.
9. Write a C program to store and display student information using structures.
10. Write a program to find total and average of N students using structures.

UNIT-2: Complexity Analysis & Sorting Techniques

1. Define Time Complexity and Space Complexity.
2. Explain Asymptotic Notations.
3. Explain Merge Sort algorithm with example.
4. Write Merge Sort program.
5. Explain Quick Sort algorithm with example.
6. Write Quick Sort program.
7. Explain Radix Sort with example.
8. Compare Quick Sort and Merge Sort.
9. Trace Merge Sort for a given array.
10. Trace Quick Sort for a given array.

UNIT-3: Linked Lists

1. Define Linked List. Explain representation of Linked List.
2. Explain advantages and disadvantages of Linked Lists.
3. Explain Singly Linked List operations.
4. Write program for insertion in Singly Linked List.
5. Write program for deletion in Singly Linked List.
6. Explain traversal operation in Linked List.
7. Explain Doubly Linked List with diagram.
8. Explain operations on Doubly Linked List.
9. Explain Circular Linked List with example.

10. Compare Arrays and Linked Lists.

UNIT-4: Stacks and Queues

1. Define Stack. Explain Stack operations.
2. Implement Stack using Arrays.
3. Implement Stack using Linked List.
4. Explain applications of Stack.
5. Define Queue. Explain Queue operations.
6. Implement Queue using Arrays.
7. Implement Queue using Linked List.
8. Explain Circular Queue with diagram.
9. Evaluate Postfix Expression using Stack.
10. Differentiate Stack and Queue.

UNIT-5: Trees and Graphs

1. Define Tree and explain tree terminology.
2. Explain types of Trees.
3. Explain Binary Tree Traversals with examples.
4. Write algorithms for Inorder, Preorder and Postorder traversals.
5. Explain Binary Search Tree (BST).
6. Explain BST insertion and deletion.
7. Define Graph and explain graph terminology.
8. Explain Adjacency Matrix and Adjacency List representations.
9. Explain Breadth First Search (BFS) algorithm.
10. Explain Depth First Search (DFS) algorithm.